

Short-Time Fourier Transform Covariance and Selection, A Feature Extraction Method for Binary Motor Imagery Classification

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Abstract—The brain-computer interfaces (BCI) technology is able to help dysfunctional people recover their motor functions. Electroencephalography (EEG) is an effective noninvasive method to construct BCI. Motor imagery (MI) paradigm can directly reflect the motor intention of user without additional stimulation equipment in EEG-BCI. The feature extraction methods are the key components to improve the accuracy of MI classification. Traditional feature extraction methods like CSP that only extract features in a single domain or two domains. In this study, we propose two novel feature selection method, short-time Fourier transform covariance and its selection method, which are aiming to extract spatial-time-frequency features simultaneously. In order to evaluate the proposed features, the BCI Competition IV Data Set IIb is employed to test the classification accuracy. By comparing the average accuracy of novel TSGSP method, the proposed method is more stable than TSGSP about 6% and the accuracy is just decrease about 0.5% at the same time. The average accuracy of 83.8% over all subjects is obtained. Superior classification performance results show that our proposed method has great potential, which is helpful for the further development and application of BCI technology for motor imaging in the field of neurorehabilitation.

I. INTRODUCTION

The brain-computer-interface (BCI) is a natural human-machine-interface (HMI) that can be used to help motor dysfunction patients interact with external auxiliary devices, such as exoskeleton robots [1], prostheses [2], and smart wheelchairs [3], so as to recover motor ability. BCI can be classified into invasive and non-invasive according to the degree of intrusion into the human body. Invasive BCI such as ECoG have good signal quality and have been proven to be effective enough for control of external equipment [4]. However, to equip invasive BCI, the participants require to have brain surgery. The risks and costs of the surgery are very high. Non-invasive BCI mainly include: fMRI, fNIR, MEG, EEG, etc. The MEG has good spatial and time resolution, but the system of MEG is very large and cannot be integrated into wearable devices. The fMRI has the best spatial resolution, but the time resolution is low. Besides, like the MEG, the fMRI system is very large too. The fNIR is portable and its signal is not

easily affected by EMG artifacts. However, its time resolution is not high, so the intent of participants can not be reflected in real time. EEG is susceptible to interference such as myoelectric artifacts, but it has a good time resolution and is suitable for constructing HMI for external devices that need to consider response time. To construct EEG-BCI, the steady-state visual evoked potentials (SSVEP) based method [5] and event related desynchronization (ERD) based motor imagery (MI) method [6] are often employed. The SSVEP-based method requires external stimulation equipment and cannot reflect the participant's intention in real time because of the participants are evoked potentials after the they have decisions by focusing on the corresponding stimulation. While the MI method can directly reflect the participant's spontaneous motor intention without additional stimulation equipment. Unfortunately, the MI is easily affected by artifacts, so how to extract effective features of EEG signals in MI is the key to construct MI-BCI with good performance.

In general, the features in spatial, time, frequency and time-frequency domain are extracted to construct MI-BCI. The Common Spatial Pattern (CSP) is the widely used spatial feature extraction method [7]. It maximizes the signals' variance of the inter classes and minimizes the variance of inner class to obtain the spatial features with the largest discrimination. As a variant of CSP, Ang et al. [8] proposed the filter bank CSP (FBCSP) method, which divides the frequency of the signal into multiple sub-bands by band-pass filter to obtain multiple CSP features along sub-bands, and select the subbands CSP with the best performance for classification, which is able to improve the classification performance. Peterson et al. [9], [10] decompose the EEG signal into sub-segments with multiple bands by band-pass filter and time windows for extracting multiple CSP features, then perform feature selection and classification at the same time, the performance of this method is more stable. Their works are aiming to extract spatial features along time and frequency dimensions, however, by using the band-pass filter the information in the frequency domain does not actually be extracted. The Fourier transform is the basic and effect tool to obtain the features in frequency domain [11], [12] and the short-time Fourier transform (STFT) is the improved tool to obtain the features in the time-frequency domain. The STFT features are widely used in MI tasks [13], [14]. Another commonly employed features in time-frequency

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domain is the wavelet transform (WT) [15]. However, these methods are not fully considering the effects of the spatial domain.

Therefore, considering that the features of EEG signals in different domains has different effects on improving the classification accuracy of the MI-BCI system, this paper combines the advantages of the CSP method in the spatial domain and the advantages of the STFT method in the time-frequency domain, and proposes two feature extraction methods which represent the spatial-time-frequency domain information at the same time, the STFT covariance method and STFT covariance selection method. The STFT covariance selection method introduces a feature selection method on the basis of STFT covariance, and finds out important frequency bands and spatial locations under different time segments. In summary, the main contributions of this article are as follows:

- A feature extraction method STFT covariance that integrates spatial-time-frequency domain information is proposed to make up for the shortcomings of traditional feature extraction methods that only extract features in a single domain or two domains.
- The novel feature selection method, the infinite feature selection, is introduced in the STFT covariance feature extraction method to improve the stability and effectiveness of proposed features.

The organization of this paper is as follows. In section II, the proposed feature extraction methods are explained in detail, the CSP and STFT feature extraction methods are also introduced to better describe the concept of proposed methods. In section III, the EEG data set, experiment process and experimental evaluation criterion are described. In section IV, the results are analyzed, and the characteristics and performance between different features are discussed. Finally, the conclusion and future work of this research are summarized in section V.

II. Methods

In this section, the commonly used spatial domain feature extraction method CSP algorithm and the time-frequency domain feature extraction method STFT are introduced first as the base of the proposed features extraction method. Then we consider all the spatial, time and the frequency domains and propose two new features extraction methods, the STFT Covariance (STFTCOV) method and the STFTCOV selection method, based on the STFT and CSP.

A. Common Spatial Pattern

The CSP algorithm is a efficient method in dealing with binary classification problem, the CSP maximize the variance of one category and minimize the variance of another category at the same time. We let the $X \in R^{N \times S}$ as the windowed data of EEG signal, in which the N and the S are the number of EEG channels and the number

of samples in the window. The CSP is to project the X to Z according to the spatial filter W :

$$Z = WX, \quad (1)$$

The spatial filter W is obtained by diagonalizing the covariance matrix of two categories simultaneously. For i -th windowed data X_i , the covariance matrix is defined as:

$$C_i = \frac{X_i X_i^T}{\text{trace}(X_i X_i^T)}, \quad (2)$$

$\text{trace}(X)$ is the function of calculating the trace of the matrix. The covariance matrix of category one $C1$, the covariance matrix of another category $C2$ and the covariance matrix of two categories are defined as follows:

$$C1 = \text{mean}(C_{\{i \in \text{category } 1\}}), \quad (3)$$

$$C2 = \text{mean}(C_{\{i \in \text{category } 2\}}), \quad (4)$$

$$C_C = C1 + C2 = F_C \psi F_C^T, \quad (5)$$

in which, F_C is the eigenvectors of eigenvalues ψ . We can construct the common eigenvectors $S1$ and $S2$ based on whitening matrix P as:

$$S1 = P(C1)P^T, S2 = P(C2)P^T, P = \psi^{-1/2} F_C^T, \quad (6)$$

then, the $S1$ and $S2$ can be decomposed into:

$$S1 = B_1 \Lambda_1 B_1^T, S2 = B_2 \Lambda_2 B_2^T, \quad (7)$$

in which, the B_1, B_2 are the eigenvectors of eigenvalues Λ_1 and Λ_2 . It can be prove that:

$$B_1 = B_2 = U, \quad \Lambda_1 + \Lambda_2 = I, \quad (8)$$

therefore, the spatial filter W can be calculated as:

$$W = U^T P. \quad (9)$$

The covariance matrix of all channels are the base of spatial filter in CSP algorithm. Hence we extract spatial features according to the covariance matrix in the subsequent subsections.

B. Short-Time Fourier Transform

The STFT is used to analyze the connection of time and frequency domain of the nonstationary signal and obtain the time-frequency features. It calculate the frequency content of a sliding short-time windowed signal. In this work, the sliding short-time window length is M , and the discrete Fourier transform (DFT) of the windowed data are employed. The intervals of time window is R samples and we also employed the overlap length L at the same time to compensate for the signal attenuation at the window edges. It is easy to find that $R = M - L$. The STFT features of i -th trial EEG signal $x(n)$ are $X_i = [F_1, F_2, \dots, F_c]$, in which, c is the channel of the EEG data, $F_j = [f_1, f_2, \dots, f_k]$ is the STFT matrix of the j -th channel. The number of rows l is set to 1024

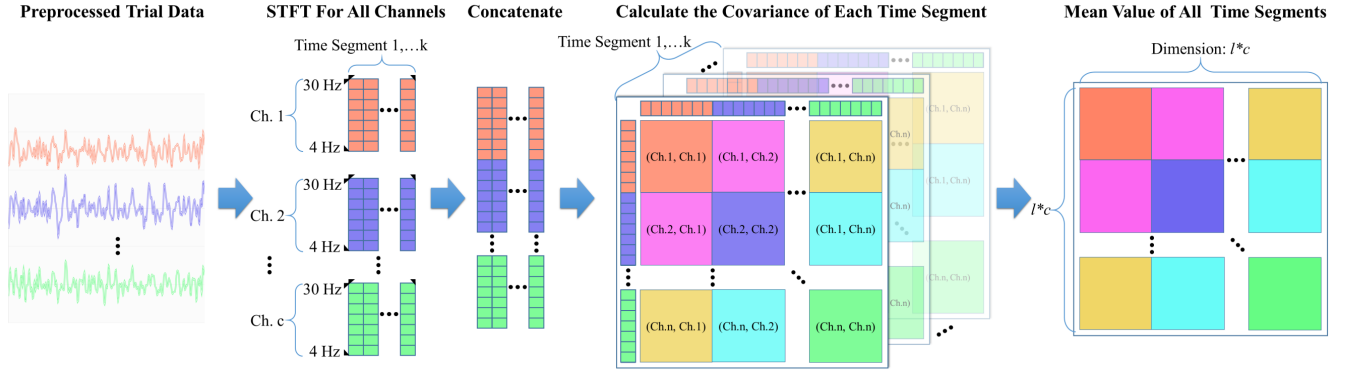


Fig. 1. The feature extraction process of STFTCOV.

points, and the $x(n)$ is padded with trailing zeros to length 1024. The number of columns k is given by

$$k = \left\lfloor \frac{N_x - L}{M - L} \right\rfloor, \quad (10)$$

where N_x is the length of the i -th trial EEG signal and the $\lfloor \cdot \rfloor$ symbols is the floor function. The m -th element of $\mathbf{f}_m$ is

$$\mathbf{f}_m = \sum_{n=-\infty}^{\infty} x(n)g(n - mR)e^{-j2\pi fn}, \quad (11)$$

in which, the $\mathbf{f}_m$ is the DFT of the windowed EEG signal centered about time mR , the $g(n)$ is the window function of length M and given as follows

$$g(n) = \frac{I_0\left(\beta\sqrt{1 - \left(\frac{n-M/2}{M/2}\right)^2}\right)}{I_0(\beta)}, \quad 0 \leq n \leq M, \quad (12)$$

where $I_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind

$$I_\nu(z) = \left(\frac{z}{2}\right)^\nu \sum_{(k=0)}^{\infty} \frac{\left(\frac{z^2}{4}\right)^k}{k!\Gamma(\nu + k + 1)}, \quad (13)$$

where ν is the order of modified Bessel function of the first kind, $\Gamma(\cdot)$ is the Gamma function

$$\Gamma(z) = \int_0^\infty e^{-t}t^{(x-1)}dt, \quad x \in \mathbb{R} > 0. \quad (14)$$

The channels c in BCI Competition IV DataSet IIb is 3, and the N_x in each trial is 750 samples. We set the window length M and overlap L are 50 and 25 samples to make the time resolution in 0.2s. Hence, the number of features in each trial are $c * l * k = 3 * 1024 * 29$. In order to decrease the number of features, we compress the power spectrum into $512 * 29$ by cubic interpolation first. Then all compressed features in three channels are summed to combine the spatial dimensions in a simple way. As last, the number of features in STFT is $512 * 29$.

C. Short-Time Fourier Transform Covariance

In STFT we employ a simple strategy to obtain the features of different channels which is to calculate the mean value of STFT features along all channels. However, from the perspective of CSP algorithm, the spatial features are significant in binary MI classification. It is necessary to organize the spatial features well. Therefore, we employ the covariance of all channels to combine the spatial features into the time-frequency features in the proposed Short-Time Fourier Transform Covariance (STFTCOV) feature extraction method. As the Section II-B, the dimensions of STFT features of i -th trial EEG signal are $\mathbf{X}_i \in \mathbb{R}^{c \times l \times k}$ before feature compression. Because of the time resolution is low in STFT, it is 0.2s, so we divide the STFT features along the time dimension. For easily understood, we re-define the STFT features of the i -th trial EEG signal $\mathbf{X}_i$:

$$\mathbf{X}_i = \begin{bmatrix} \mathbf{f}_{1,1} & \mathbf{f}_{1,2} & \cdots & \mathbf{f}_{1,k} \\ \mathbf{f}_{2,1} & \mathbf{f}_{2,2} & \cdots & \mathbf{f}_{2,k} \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{f}_{c,1} & \mathbf{f}_{c,2} & \cdots & \mathbf{f}_{c,k} \\ \underbrace{\mathbf{d}_1}_{\mathbf{d}_1} & \underbrace{\mathbf{d}_2}_{\mathbf{d}_2} & \cdots & \underbrace{\mathbf{d}_k}_{\mathbf{d}_k} \end{bmatrix}, \quad (15)$$

where the $\mathbf{f}_{p,q} \in \mathbb{R}^{l \times 1}$ is the power spectrum vector of p -th ($p \in \{1, 2, \dots, c\}$) channel and q -th ($q \in \{1, 2, \dots, k\}$) time segment. The $\mathbf{d}_q \in \mathbb{R}^{(c \times l) \times 1}$ is the vector of q -th time segment. Hence, the covariance of all spatial channels are as follows:

$$\mathbf{\Sigma}_q = \frac{\mathbf{d}_q \mathbf{d}_q^T}{\text{trace}(\mathbf{d}_q \mathbf{d}_q^T)}, \quad (16)$$

in which, $\mathbf{\Sigma}_q \in \mathbb{R}^{(c \times l) \times (c \times l)}$ is the STFT covariance features matrix of q -th time segment. The STFTCOV features is given by $\mathbf{C}_i = [\mathbf{\Sigma}_1, \mathbf{\Sigma}_2, \dots, \mathbf{\Sigma}_k]$. In order to decrease the number of features, we compress the features along time dimension by obtain the mean of all the $\mathbf{\Sigma}_q$ as $\mathbf{C}_{i,out} = \text{mean}(\mathbf{\Sigma}_1, \mathbf{\Sigma}_2, \dots, \mathbf{\Sigma}_k) \in \mathbb{R}^{(c \times l) \times (c \times l)}$. The whole process of STFTCOV method is shown in Fig. 1.

In this work, we resize the dimension of $\mathbf{C}_{i,out}$ to $\mathbb{R}^{50 \times 50}$ to accelerate the computation time. The di-

mension 50×50 is a empirical value which is able to compromise accuracy and calculation speed.

D. Short-Time Fourier Transform Covariance Selection

In the STFTCOV, we compress the features along the time dimension by calculate the mean value of each covariance matrix. Obviously this method will filter out some useful details of each time segment. Hence, we introduce the Infinite feature selection [16] (Inf-FS) method to select useful covariance features in each time segment to deal with the problems of ommiting useful details in STFTCOV. The Inf-FS method models the parametric searching problem into the undirected fully-connected weighted graph $G = \{V, E\}$, in which $V = \{v_1, v_2, \dots, v_n\}$ representing the feature distribution of a n -dimension feature set $F = \{f_1, f_2, \dots, f_n\}$. The E is the edge set that represents the connection between a pair of nodes. If the graph G is represented by the adjacency matrix A , the element of adjacency matrix $A(i, j)$, $1 \leq i, j \leq n$ can be given by

$$A(i, j) = \phi_s(v_i, v_j) = s_i s_j, \quad (17)$$

where s_i is the linear summation factor of i -th features that is represented as follows

$$s_i = \alpha_1 h_i + \alpha_2 m_i + \alpha_3 \sigma_i, \quad \alpha_1 + \alpha_2 + \alpha_3 = 1 \quad (18)$$

where h_i , m_i and σ_i are Fisher discriminant criterion, mutual information and standard deviation respectively to evaluate the redundancy, correlation and divergence between features. Therefore, the weight $R_l(i, j)$ of any path length l , that is the any combinations of feature set, is as follows

$$\pi_\gamma = \prod_{k=0}^{l-1} A(v_k, v_{k+1}), \quad (19)$$

$$R_l(i, j) = \sum_{\gamma \in \mathbb{P}_{i,j}^l} \pi_\gamma = A^l(i, j), \quad (20)$$

in which, π_γ is the weight of all feature pairs on the path from feature i to feature j . Because of not only one paths from one feature to another, the $\mathbb{P}_{i,j}^l$ is defined as the path set including all paths which length is l . So the $R_l(i, j)$ include the weight of all the paths. In the perspective of Matrix algebra, go through all the paths is equivalent to evaluate each feature multi-times. So the $R_l(i, j)$ is replaced by $A^l(i, j)$. Therefore, to evaluate the importance $c(i)$ of i -th feature is equivalent to consider all the possible combination of feature set by making the path length $l \rightarrow \infty$, that is

$$c(i) = \sum_{l=1}^{\infty} \left(\sum_{j \in V} R_l(i, j) \right) = \left[\sum_{l=1}^{\infty} A^l \right]_i = [C e]_i \quad (21)$$

$$C = \sum_{l=1}^{\infty} A^l, \quad (22)$$

where e is a ones vector. Because of the A^l is divergent, the $c(i)$ cannot be obtained. However, if we define the isomorphic form $\check{c}(i)$ as

$$\check{c}(i) = \sum_{l=1}^{\infty} \sum_{j \in V} r^l R_l(i, j), \quad (23)$$

when the $r = 0.9/\rho(A)$, the $\check{c}(i)$ will be convergence. The $\rho(A)$ is the maximum eigenvalue of matrix A . Hence, we can obtain the importance and the order of each features by calculating the redundancy, correlation and divergence criterion.

In this work, we only employ the correlation criterion of all features, so we set the $\alpha_1 = 0$, $\alpha_2 = 1$ and $\alpha_3 = 0$. The number of selected features in each time segment are same and the number is optimized by the particle swarm optimization (PSO) method for each subject. In order to compress the feature further, we select the features after combined time segments by Inf-FS again to obtain the final features. The process of short-time Fourier transform covariance selection (STFTCOVSELECT) is shown in Fig. 2.

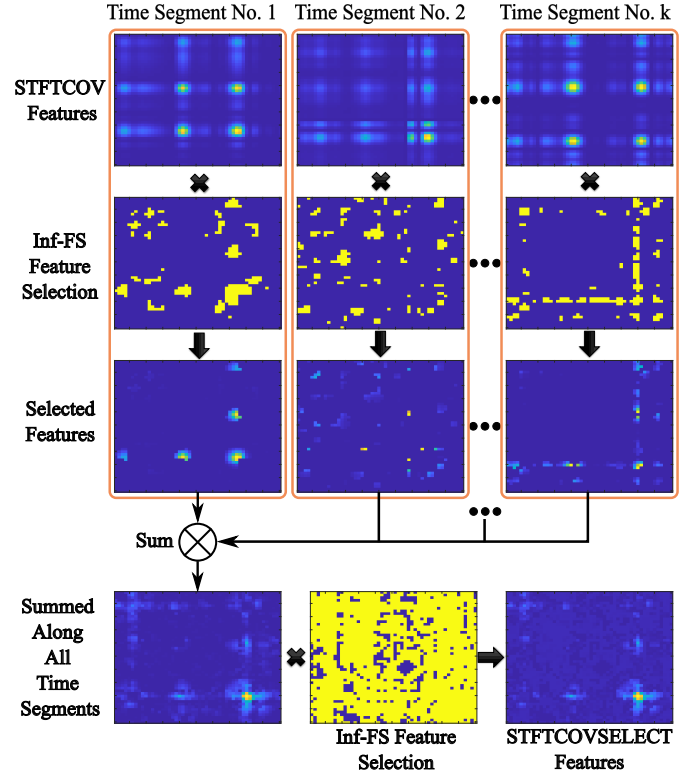


Fig. 2. The feature extraction process of STFTCOVSELECT.

III. Experiments

In this section, the platform of the experiments are introduced first, then the datasets and experiments protocol of the experiments are proposed. At last the evaluation criterion of our experiments are described.

A. Platform and Toolbox

The proposed methods and the experiments are all implemented in Matlab R2020b (MathWorks, USA). The EEGLAB toolbox on Matlab are employed to preprocess the EEG-based MI dataset. The STFT features are extracted by using the Matlab Signal Processing Toolbox. The SVM classifier of our experiments is constructed based on the Matlab Machine Learning Toolbox. The PSO is implemented based on the Matlab Optimizaiton Toolbox. All the experiments are evaluated in a computer with Intel(R) Core(TM) i7-6700 3.40-GHz processor and 16GB of memory.

B. Experiments Protocol

In the experiments, the BCI Competition IV DataSeT IIb is employed to validate the proposed methods. Only the third training sessions of the dataset are used in this study, i.e., "B0103T", "B0203T", ..., "B0903T". Nine subjects are contained in this dataset, all the subjects are right-handed. The EEG signals of C3, C4 and Cz channels of International 10-20 system were recorded at 250 Hz for left hand and right hand MI tasks. For each subject, a total of 160 trials of MI task experiments with visual feedback were recorded (80 for left-hand MI and 80 right-hand MI). For each trial, there is a 3.5s preparation stage, followed by a 4.5s MI visual cue, informing the subjects that they need to perform a left-handed or right-handed MI task, and end with a random rest time of 1-2s. More details about the dataset could be found in <http://www.bci.de/competition/iv/>.

The experiment process is shown in the Fig. 3. Because of the parameters optimization is needed, the experiments is divided into optimization process and test process. In the optimization process, we optimize two parameters by classification accuracy: feature number of each time band and feature number of different subjects. Then, in the test process, these two parameters are fixed to test the effects of the proposed method. An other diffences between the optimization process and test process are the train set and test set ratio.

The ratio of train set and the test set are 0.7 : 0.3 in the optimization process and 0.9 : 0.1 in the test process. The reason why we choose different ratio in the optimization process and test process are that we want to test the optimized parameters are still effective when the train set and test set are changed. For every subject ($Subjects = 1, 2, \dots, 9$) in the optimization process and test process, we randomly select the train set and test set for 10 times to validate the performance of proposed method.

C. Evaluation Criterion

In this work, we define the average classification accuracy of different subjects ($ACAS_i$, $i = 1, 2, \dots, N$, in this work, the $N = 9$) and average classification accuracy of the dataset (ACA) to evaluate the performace of the proposed method. For the i -th subject, the $ACAS_i$ is the average classification accuracy of all repeat times M :

$$ACAS_i = \frac{CA_1 + CA_2 + \dots + CA_M}{M}, \quad (24)$$

where CA_j , $j = 1, 2, \dots, M$ is the classification accuracy of the j -th experiment. For all the N subjects in the dataset, the ACA is defined as:

$$ACA = \frac{ACAS_1 + ACAS_2 + \dots + ACAS_N}{N}. \quad (25)$$

IV. Results

In order to test the performance of the proposed STFTCOV and STFTCOVSELECT features, we compare the $ACAS_i$ and ACA of CSP, STFT, STFTCOV and STFTCOVSELECT features by SVM, the results are shown in Table I. We also compare the proposed method to novel TSGSP [17] method to verify its effectiveness. From the Table I, it is obviously to find that the STFTCOVSELECT obtain the best performance of the employed four methods. The STFT and the STFTCOV get terrible performance, this is because the "Curse of dimensionality". The dimension of STFT and STFTCOV are $\mathbb{R}^{512 \times 29}$ and $\mathbb{R}^{50 \times 50}$. It is too large for the SVM classifier. Therefore, it is necessary to employ feature selection method in these feature extraction methods.

Comparing the STFTCOVSELECT to TSGSP, the ACA of STFTCOVSELECT is lower than TSGSP about 0.5%, however, the standard deviation of STFTCOVSELECT is lower than TSGSP about 1%. From the perspective of the rate of increase/decrease, we can find the accuracy decrease rate of STFTCOVSELECT from TSGSP is about $((83.8 - 84.3) / \max(83.8, 84.3)) \times 100\% = -0.6\%$ and the standard deviation decrease rate is about $((14 - 15) / \max(14, 15)) \times 100\% = -6.7\%$. This proves that the proposed method is more stable than TSGSP about 6% and the accuracy is just decrease about 0.5% at the same time. When we check the accuracy of each subject, it can be found that the STFTCOVSELECT have better performance in dataset of subject 2, 3, 5, 8, and 9. Especially in dataset 303, the proposed method increase the classification accuracy

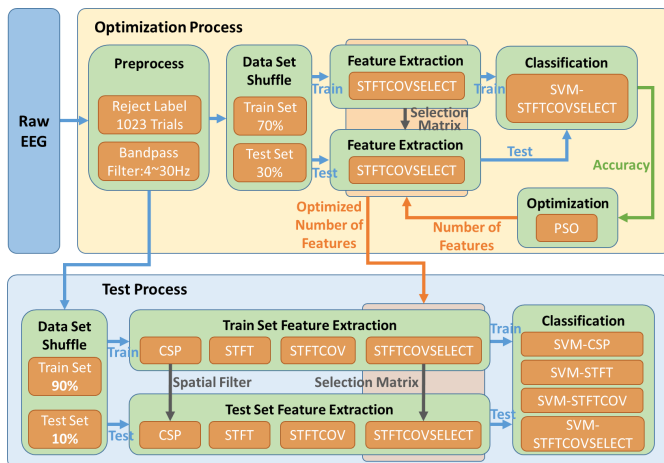


Fig. 3. The experiment process.

TABLE I

The average classification accuracy of different subjects (%). Bold shows the highest average accuracy of each subject. Note: FE-Feature Extraction, TSGSP-Temporally Constrained Sparse Group Spatial Pattern proposed by Y. Zhang et. al.

Features	103	203	303	403	503	603	703	803	903	ACA
CSP	78.9	64.4	44.2	98.0	85.8	72.4	86.3	85	85.1	77.8±15.8
STFT	62.2	55.9	46.7	58.9	51.2	51.0	58.3	67.2	55.6	56.3±6.2
STFTCOV	51.4	51.3	52.8	50.0	51.2	51.2	51.2	50.0	51.2	51.1±0.8
STFTCOVSELECT	77.5	63.8	61.7	98.8	96.4	80.0	87.9	94.0	94.3	83.8±14
TSGSP [17]	84.0	62.6	56.3	99.4	94.8	83.8	94.1	93.3	90.1	84.3±15

about 5.4% and the accuracy increase rate of dataset 303 is about $((61.7 - 56.3)/\max(61.7, 56.3)) \times 100\% = 8.8\%$. However, in dataset 103 and 703, the proposed method decrease the classification accuracy about 6.5% and 6.2% respectively. To find out the reason why the accuracy of these two datasets have lower performance, we check the results of optimization process in which the ratio of train set and test set are 0.7 : 0.3. The average accuracy of dataset 103 and 703 in optimization process is 82.2% and 90.5%. This suggest that some trials in dataset 103 and 703 have different feature distribution. When these 'odd' trials are diluted by other 'fine' trials, the accuracy of these dataset is increasing. Hence, it is necessary to find out and reject these 'odd' trials in the future work to increase the accuracy and stability further.

V. Discussion

In this paper, we propose two novel feature selection method STFTCOV and STFTSELCT, which are aiming to extract spatial-time-frequency features simultaneously. The STFTCOV extraction method pull in the merit of CSP algorithm, and obtain the spatial feature by calculating the covariance of all channels. The STFTCOVSELECT extraction method introduce the Inf-FS feature selection method into STFTCOV to find out the important frequency features along all time segments. Thus, we obtain the features that have good classification accuracy and stability.

The proposed features have good performance, but we only employ the simply accumulation along all time segments in this paper. The time segment selection method need to be conducted in the future work. We also find that there are some different feature distribution trials in all the dataset, it is necessary to remove these trials to increase the classification accuracy and stability. The STFTCOV and STFTCOVSELECT feature extraction method is memory consuming, so some feature compression method need to be researched for future research.

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